

Gaussian Splatting

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CSE 575 - Fall 2025

Abstract

Objective:

The goal of this project is to

- Implement the 3D Gaussian Splatting algorithm to successfully reconstruct a high-quality 3D visualization of a scene/image from given dataset of 2D images
- If possible, design an interactive tool utilizing the project as a backend and modify the Gaussian parameters in real-time / incorporate dynamics to the imagery (lighting based on angle, time component to depict scene over a day, etc.)

Motivation:

Gaussian splatting represents a modern approach to generating 3D photorealistic scenery. Neural Radiance Fields (NeRFs), while effective in content creation, are too costly computationally (utilizes a deep neural network) and are slow to train and render scenery. Gaussian Splatting represents a solution that addresses the issues of NeRF, drastically speeding up training and rendering.

Our motivation is to gain an implementation-level understanding of this technology. This could be utilized in the 3D animation industry in the near future for movies, simulations, virtual reality, etc. We also have the opportunity to learn how this type of scene representation overcomes the limitations of previous implementations, and how 3D computer generated scenery has evolved over time.

Expected Outcome:

We hope to at least have a working prototype of a model capable of taking in a satisfactory 2D image set, and producing a realistic depiction of the actual 3D scene the image set was based around. This model should also perform at decent performance, at least surpassing a NeRF implementation

Execution Plan

Steps:

- Conduct research on university and other open-source implementations of 3DGS (3D Gaussian Splatting). Setup the proper development environment
- Capture a custom image dataset from a static scene. Can be done manually, or use online datasets. Process imagery through COLMAP to generate data to train on
- Implement 3DGS model
 - Implement data loading, 3D Gaussian representation, differentiable rasterizer, training loop
- Report findings in a comprehensive research paper, detailing all previous steps and describing how and why we approached the project

Workload Distribution:

The distribution is tentative and will be modified in the future as we progress on the project and evaluate next steps. Here is an initial outlook of what it could look like:

- Research: Anish and Neha
- Gather data: Cephus and Satyam
- Implementation: All (divide later)
- Report: Mawadda

Expected Challenges:

It's a bit difficult to know for certain what challenges we will face in the future as we develop this project.

Some potential anticipated difficulties include:

- Developing high quality image set to train on
- Training the model to produce realistic 3D imagery
- Debugging the model
- Understanding the low-level linear algebra of the implementation
- Hardware requirements (could be alleviated using Sol)

We hope to be able to generate solutions to these problems as we gain more experience developing. Some ideas to alleviate the concerns we have right now are:

- Utilizing external software (ex. COLMAP) to process images as better training data
- Keeping logs of the model's output at every stage, as well as visualize at each step whenever possible to see how the model works
- Isolate variables / change one hyperparameter at a time

Evaluation Plan

Outcome Evaluation:

We will evaluate the outcome of the project by first assessing that it can successfully produce a 3D render given an 2D image set. The metric of success comes from how accurately the model's produced 3D image replicates the setting of the 2D image set. We will also compare how our model performs to other implementations (Blender has a Gaussian Splatting add-on that we will compare with). We will also evaluate efficiency by monitoring the runtime performance and memory usage. A successful outcome can be determined through measuring the consistency across multiple datasets when handling more complex image sets.

Teammate Evaluation:

We aim to evaluate each other's performance simply through clear effort and contribution.

Communication about how each member's tasks are going, if there are any difficulties in the tasks and asking for help, are going to be expected from each member to facilitate development. Additionally, simply being able to complete or at least contribute meaningfully to the tasks assigned is all that would be needed out of every member.